NR2DT-04: Step 4 — Translate

> **Series:** NR2DT | **Notebook:** 4 of 10 | **Created:** April 2026 | **Last Updated:** 04/14/2026

Overview

Goal of this step: convert every NRQL query in the inventory to DQL. Auto-translate HIGH; review MEDIUM; hand-translate LOW. Resolve every gap before any artifact migrates.

Procedural — see **NRLC-02** for the compiler architecture, the 292 patterns, and confidence theory.

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Prerequisites

Requirement	Details
Audience	Migration lead + assigned engineer for this step

Requirement	Details
Completed	NR2DT-03 — Design
Format	Procedural step — use as a runbook; defer to NRLC for depth
NRLC deep dives	NRLC-02 (NRQL→DQL Translation deep dive)

1. What You'll Produce

Artifact	Format	Purpose
`translated-queries.csv`	CSV	All queries with final DQL + confidence
`manual-translations.md`	Markdown	LOW-confidence hand-translations with rationale
`dql-validation-report.md`	Markdown	Tier 1 syntax validation results
`translation-coverage.md`	Markdown	% of queries translated; remaining gaps

2. Run the Translator

```
# Re-run translation against the latest inventory (Step 1 produced it)
python3 migrate.py compile --file inventory/all-nrql.txt \
    --output translated-queries.csv --report
```

Output columns: `source_nrql, translated_dql, confidence, score, notes, warnings, fixes`.

Sort by confidence ascending — attack LOW first, then MEDIUM, save HIGH for sample-validate.

3. Review MEDIUM Confidence

MEDIUM means the translation is structurally correct but has edge cases worth verifying. Common review checklist:

Pattern	What to verify
`COMPARE WITH`	Time alignment correct; append produces the expected shape
Subquery via `lookup`	Lookup target exists as queryable Grail data
`SLIDE BY`	Sliding window semantics preserved
Nested `if()` in `FACET CASES`	Nested DQL emits the right structure
Custom event types	Source mapping (FROM → fetch) correct

Process: mark each MEDIUM as APPROVED or NEEDS-WORK in the CSV. NEEDS-WORK becomes a hand-translation.

4. Hand-Translate LOW Confidence

LOW translations contain TODO markers or are missing entirely. Common causes:

Cause	Action
`apdex(field, t:N)`	Configure DT Apdex via custom DQL or use Davis SLO
`funnel(...)`	Reformulate as nested `append` / `lookup` chain
`cohort(...)`	No DT equivalent — reformulate the requirement
Custom event type with no source mapping	Add ingest mapping via OpenPipeline first

Document each hand-translation in `manual-translations.md` with the source NRQL, the new DQL, and the rationale.

If a query can't be translated and the requirement can't be reformulated, document it as **deferred** — it joins the gap analysis from Step 1.

5. Validate Translated DQL

Tier 1 syntax validation — confirm every DQL parses against the target tenant:

```
python3 migrate.py validate --file translated-queries.csv \
    --tier syntax
```

Failures attempt auto-fix (`DQLFixer`) and re-validate. Final report shows pass / fail / fixed-then-passed counts.

Tier 2 tenant validation (referenced metrics/entities/buckets exist) runs in Step 8 before each component imports.

6. Step Exit Criteria

G4 — Translation Complete

- [ ] Every query has a final DQL (auto, reviewed-MEDIUM, or hand-translated)
- [ ] All translations pass Tier 1 syntax validation
- [ ] LOW-confidence count is 0 (all triaged)
- [ ] `manual-translations.md` documents every LOW resolution
- [ ] Deferred queries documented with deferred-rationale

Next step: NR2DT-05 — Migrate Dashboards & Alerts.
